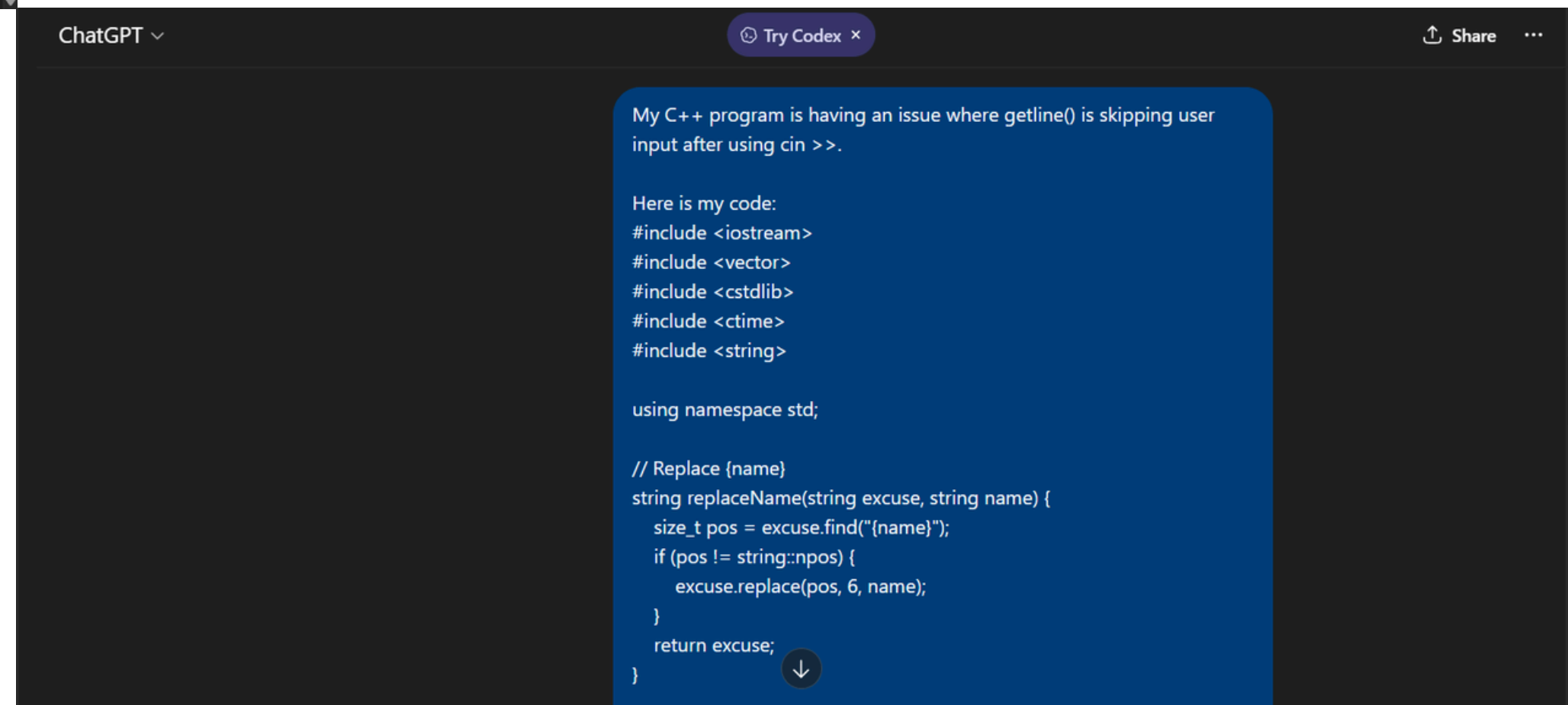
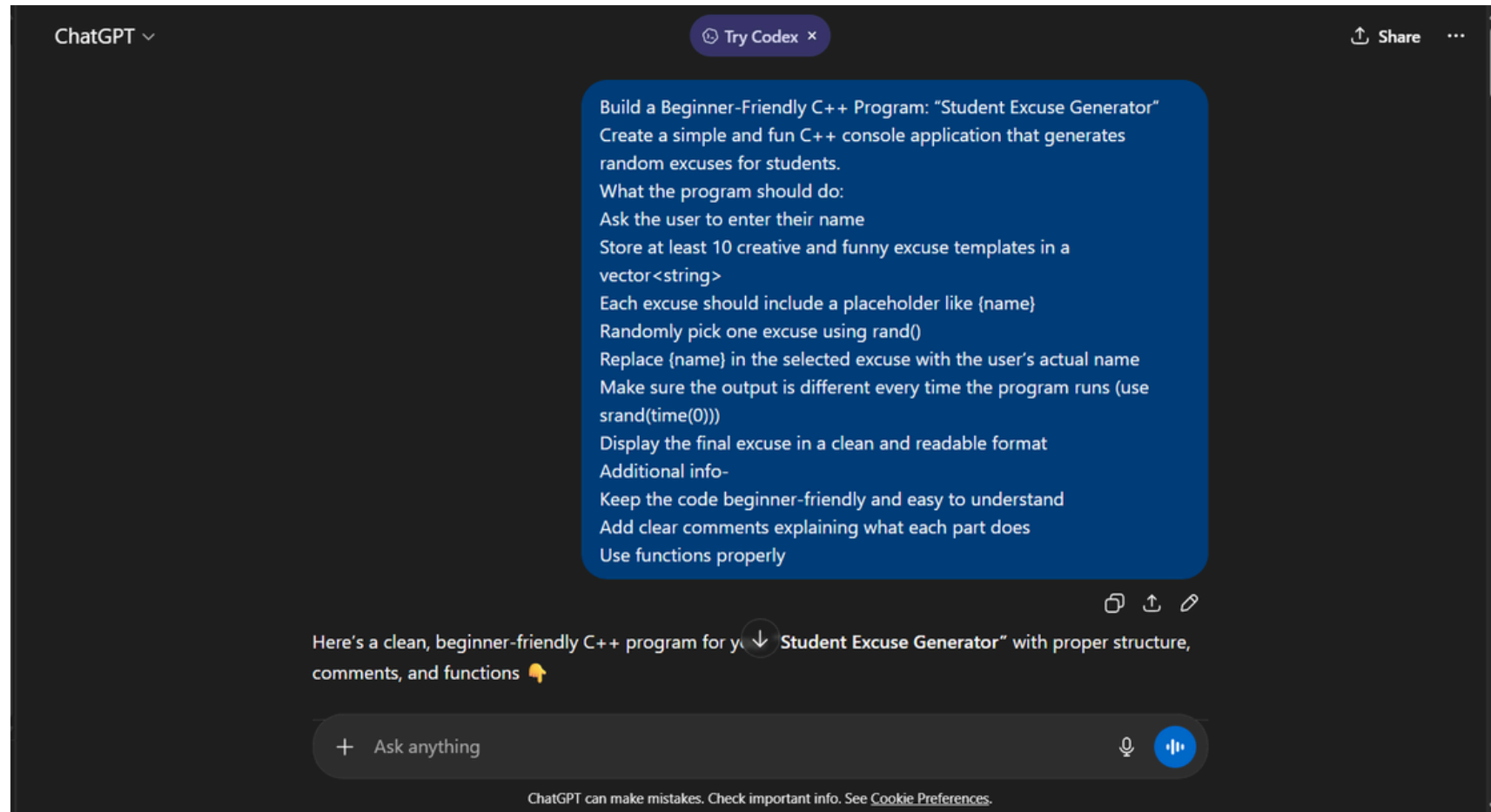


FOCP-II

**THIS PROGRAM GENERATES RANDOM EXCUSES
FOR STUDENTS USING C++. IT TAKES USER
INPUT AND DISPLAYS A RANDOMLY SELECTED
EXCUSE BY REPLACING PLACEHOLDERS
DYNAMICALLY.**

PROMPTS USED



```
// Generate random excuse
string generateExcuse(string name) {
    vector<string> excuses = {
        "{name} forgot the homework because a monkey stole the
        notebook.",
        "{name} was busy saving the world from aliens last night.",
        "{name}'s dog ate not only the homework but also the pen.",
        "{name} accidentally submitted the homework to another
        dimension.",
        "{name} was practicing magic and the homework disappeared.",
        "{name}'s internet turned into a potato and stopped working.",
        "{name} had to help grandma fight a ninja invasion.",
        "{name} thought today was Sunday... turns out it wasn't.",
        "{name} got stuck in a time loop and couldn't finish it.",
        "{name}'s brain needed a software update overnight."
    };

    int randomIndex = rand() % excuses.size();
    return replaceName(excuses[randomIndex], name);
}
```

```
int main() {
    srand(time(0));

    string name;
    int dummy;

    cout << "=== Student Excuse Generator ===" << endl;

    cout << "Enter any number: ";
    cin >> dummy;

    cout << "Enter your name: ";
    getline(cin, name);

    string finalExcuse = generateExcuse(name);

    cout << "\nYour Excuse:\n";
    cout << finalExcuse << endl;

    return 0;
}
```

```
return 0;
}
```

The problem is that it does not wait for name input and skips it.

Please fix the issue and explain why it happens.

This is a very common C++ input bug — and your code is actually almost correct. The issue isn't `getline()` itself, it's how `cin >>` behaves before it.

✖ Why the problem happens

When you use:

```
C++
```

```
cin >> dummy;
```

- It reads the number (e.g., 5)
- BUT it leaves the newline character (`\n`) in the input buffer

+ Ask anything



BUG DESCRIPTION- getline() Skipping Input

The program was not taking the user's name input correctly. After entering a number using `cin >>`, the program skipped the `getline()` input and directly moved ahead

 Why the problem happens

When you use:

```
</> C++  
cin >> dummy;
```

- It reads the number (e.g., `5`)
- BUT it leaves the newline character (`\n`) in the input buffer

Then this runs:

```
</> C++  
getline(cin, name);
```

- `getline()` sees that leftover `\n`
- It thinks: "Oh, input already ended"
- So it immediately returns an empty string without waiting

👉 That's why your input gets skipped.

Bug Fix Explanation: getline() Input Issue

Solution:

To fix this, cin.ignore() was added after cin >> to clear the input buffer. This ensures that getline() works correctly and waits for user input.

How It Works:

The cin.ignore() function discards the leftover newline character from the buffer, allowing getline() to read the intended input properly.

BEFORE

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code + - [ ] [ ] ... | [ ] [ ] X

PS C:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT> cd "c:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT\" ; if ($?) { g++ Studentexcusegenerator.cpp -o Studentexcusegenerator } ; if ($?) { .\Studentexcusegenerator }
• === Student Excuse Generator ===
Enter any number: 5
Enter your name:
Your Excuse:
    was busy saving the world from aliens last night.
• PS C:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT> cd "c:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT\" ; if ($?) { g++ Studentexcusegenerator.cpp -o Studentexcusegenerator } ; if ($?) { .\Studentexcusegenerator }
=== Student Excuse Generator ===
Enter any number: 3
Enter your name:
Your Excuse:
    was busy saving the world from aliens last night.
○ PS C:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT> 
```

AFTER

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code + - [ ] [ ] ... | [ ] [ ] X

PS C:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT> cd "c:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT\" ; if ($?) { g++ Studentexcusegenerator.cpp -o Studentexcusegenerator } ; if ($?) { .\Studentexcusegenerator }
• === Student Excuse Generator ===
Enter any number: 4
Enter your name: ANSH

Your Excuse:
    ANSH's brain needed a software update overnight.
PS C:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT> cd "c:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT\" ; if ($?) { g++ Studentexcusegenerator.cpp -o Studentexcusegenerator } ; if ($?) { .\Studentexcusegenerator }
• === Student Excuse Generator ===
Enter any number: 2
Enter your name: RAMAN

Your Excuse:
    RAMAN got stuck in a time loop and couldn't finish it.
• PS C:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT> cd "c:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT\" ; if ($?) { g++ Studentexcusegenerator.cpp -o Studentexcusegenerator } ; if ($?) { .\Studentexcusegenerator }
=== Student Excuse Generator ===
Enter any number: 9
Enter your name: ARYAN

Your Excuse:
    ARYAN was busy saving the world from aliens last night.
○ PS C:\Users\Ansh Kumar\Desktop\FOCP II- ASSIGNMENT> 
```

Conclusion

This project successfully developed a C++ Student Excuse Generator using basic concepts like vectors, functions, and randomization. The program takes user input and generates dynamic excuses.

Several bugs were identified and fixed, which improved the program's accuracy and reliability. Overall, this project helped in understanding core programming concepts and debugging techniques.